

Disproof of Conjecture 00000009614: equal Perron eigenvalues do not force isomorphic dimension groups

jilint777

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Abstract

Conjecture 00000009614 asserts that the dimension groups of two mixing shifts of finite type are order-isomorphic *if and only if* their Perron eigenvalues λ_A, λ_B generate the same real number field with agreeing unit group cosets. We refute the “if” direction. The full 4-shift, $A = [4]$, and the edge shift of $B = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ are mixing and have the same Perron eigenvalue $\lambda_A = \lambda_B = 4$, so every reading of the right-hand side holds. But $\Delta_A \cong \mathbb{Z}[1/4]$ has rank 1, while Δ_B has rank 2. So the two dimension groups are not even isomorphic as abstract groups. The disproof is formalized in Lean 4 (core library only).

1 The conjecture and the clause we refute

Definition: The dimension group of an SFT is the ordered $K_0(\mathbb{Z}^n A, \mathbb{Z}_+^n)$, whose ordered isomorphism type is a flow invariant. *Conjecture:* The dimension groups of two mixing SFTs are order-isomorphic if and only if the Perron eigenvalues λ_A and λ_B generate the same real algebraic number field with agreeing unit group cosets; explicit examples of equal entropy over different fields are given by an explicit criterion for coprime Perron numbers.

The Chinese version says the same. The conjecture is a biconditional. We refute the implication

$$[\lambda_A, \lambda_B \text{ generate the same field with agreeing unit cosets}] \implies (\Delta_A, \Delta_A^+) \cong (\Delta_B, \Delta_B^+), \quad (1)$$

which makes the biconditional, and hence the conjecture, false. We do not use the side remark about flow invariance, and we do not need the final clause about “explicit examples”.

2 Definitions

Let M be an $n \times n$ nonnegative integer matrix and X_M its edge shift. We use row vectors, as in Lind–Marcus [1].

Definition 1 (Dimension group). $\Delta_M = \varinjlim (\mathbb{Z}^n \xrightarrow{M} \mathbb{Z}^n \xrightarrow{M} \cdots)$. Concretely, its elements are classes $[v, k]$ of pairs $(v, k) \in \mathbb{Z}^n \times \mathbb{N}$, where

$$(v, k) \sim (w, l) \iff vM^{l+t} = wM^{k+t} \text{ for some } t \geq 0$$

(think of $[v, k]$ as “ vM^{-k} ”). Addition is $[u, k] + [v, l] = [uM^l + vM^k, k + l]$, the zero is $[0, 0]$, and $-[v, k] = [-v, k]$. The positive cone is $\Delta_M^+ = \{[v, k] : vM^t \in \mathbb{Z}_{\geq 0}^n \text{ for some } t\}$. The shift map is $\delta_M[v, k] = [vM, k]$.

This is Krieger's dimension group, i.e. the ordered K_0 in the conjecture. In [1, §7.5] the same group is realized inside $\mathbb{Q}^n$: if R_M is the eventual range of M (the image of $\mathbb{Q}^n$ under M^n), then $\Delta_M \cong \{x \in R_M : xM^k \in \mathbb{Z}^n \text{ for some } k\}$ via $[v, k] \mapsto vM^n(M|_{R_M})^{-n-k}$. We use the direct limit, and in Section 5 we repeat the argument in the eventual-range model, so the result does not depend on which model is used. Both of our matrices are symmetric, so row and column conventions (that is, Δ_M versus Δ_{M^T}) agree.

Lemma 2. *$\sim$ is an equivalence relation. Addition and negation are well defined, and Δ_M is an abelian group.*

Proof. Write $\pi_t(v) = vM^t$. Then $\pi_{s+t} = \pi_t \circ \pi_s$ and each π_t is additive. Reflexivity and symmetry are clear. For transitivity, suppose $uM^{l+t_1} = vM^{k+t_1}$ and $vM^{m+t_2} = wM^{l+t_2}$. Then with $t = l + t_1 + t_2$,

$$uM^{m+t} = (uM^{l+t_1})M^{m+t_2} = vM^{k+t_1+m+t_2} = (vM^{m+t_2})M^{k+t_1} = wM^{k+t}.$$

If $(u, k) \sim (u', k')$ via t , then $(uM^l + vM^k, k + l) \sim (u'M^l + vM^{k'}, k' + l)$ via the same t . Both sides become $uM^{k'+t+2l} + vM^{k+k'+l+t}$ after multiplying by $M^{k'+l+t}$, respectively M^{k+l+t} . The sum is symmetric up to $\sim$, so it is also well defined in the second argument. Negation is similar. Two identities make the group axioms immediate: $[v, k] = [vM^s, k + s]$, and $[u, k] + [v, k] = [u + v, k]$. With these, any finitely many elements can be brought to a common level k , where addition is addition in $\mathbb{Z}^n$. $\square$

Definition 3. An *order isomorphism* $(\Delta_M, \Delta_M^+) \cong (\Delta_N, \Delta_N^+)$ is a bijective additive map f with $f(\Delta_M^+) = \Delta_N^+$. A *group isomorphism* is a bijective additive map. An isomorphism of dimension triples also requires $f\delta_M = \delta_N f$.

Triple isomorphism implies order isomorphism, which implies group isomorphism. We show that *no group isomorphism* exists, which rules out all three.

A matrix $M \geq 0$ is *primitive* if some power M^k ($k \geq 1$) is entrywise positive. For an essential matrix, X_M is mixing if and only if M is primitive [1, §4.5]. The *Perron eigenvalue* λ_M is the spectral radius. By the Perron–Frobenius theorem it is the unique eigenvalue with a positive eigenvector, and $h(X_M) = \log \lambda_M$ [1, §4.2–4.4].

3 The example

Let

$$A = \begin{pmatrix} 4 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}.$$

X_A is the full 4-shift.

Proposition 4. *A and B are primitive, so X_A and X_B are mixing SFTs. Also $\lambda_A = \lambda_B = 4$, and $h(X_A) = h(X_B) = \log 4$.*

Proof. Both matrices are entrywise positive. $(1)A = 4(1)$ and $(1,1)B = 4(1,1)$ with positive eigenvectors. Also $(1, -1)B = 2(1, -1)$, so $\text{spec } B = \{4, 2\}$ and $\chi_B(x) = x^2 - 6x + 8 = (x-4)(x-2)$, with spectral radius 4. The number of paths of length k is $\sum_{i,j} (A^k)_{ij} = 4^k$ and $\sum_{i,j} (B^k)_{ij} = 2 \cdot 4^k$, because B has all column sums equal to 4. $\square$

Every reading of the right-hand side holds. The right-hand side of (1) is a condition on the pair (λ_A, λ_B) . Here $\lambda_A = \lambda_B = 4$, so the condition holds under every interpretation that is satisfied by two equal numbers. In detail:

- “Generate the same field”: $\mathbb{Q}(\lambda_A) = \mathbb{Q}(4) = \mathbb{Q} = \mathbb{Q}(\lambda_B)$.
- “Agreeing unit group cosets”: with $K = \mathbb{Q}$ and unit group $U_K = \mathcal{O}_K^\times = \{\pm 1\}$, every reasonable reading holds trivially for equal numbers. This includes $\lambda_A U_K = \lambda_B U_K$, $\lambda_A/\lambda_B \in U_K$, the classes of λ_A, λ_B in $K^\times/U_K$ agreeing, and the same with U_K replaced by any subgroup (units of an order, totally positive units, ...).
- A stronger reading that also compares the Perron eigenvectors (the ideal class of the module spanned by the eigenvector entries, which is relevant for shift equivalence when the characteristic polynomial is irreducible) is still satisfied. The normalized eigenvectors (1) and (1, 1) both have entries generating the unit ideal of $\mathbb{Z}$, so both classes are trivial.
- Equal entropy: $h(X_A) = h(X_B) = \log 4$.

In the Lean formalization this is a hypothesis: the right-hand side is an *arbitrary* relation Cond with $\text{Cond}(x, x)$ for all x .

4 The dimension groups have different ranks

Write $a \cdot x$ for the a -fold sum ($a \in \mathbb{Z}$). Two elements x, y of an abelian group are *independent* if $ax + by = 0$ with $a, b \in \mathbb{Z}$ forces $a = b = 0$.

Lemma 5 (Δ_A has rank one). *Any two elements of Δ_A are dependent. More generally this holds for any 1×1 matrix M .*

Proof. Let $p = [u, k]$, $q = [w, l]$ with $u, w \in \mathbb{Z}$. Lift both to level $K = k + l$: $p = [\mu, K]$, $q = [\nu, K]$ with $\mu = uM^l$, $\nu = wM^k \in \mathbb{Z}$. At a common level, integer combinations are computed on numerators: $ap + bq = [a\mu + b\nu, K]$. If $\mu = 0$ then $1 \cdot p + 0 \cdot q = 0$. Otherwise $(a, b) = (\nu, -\mu) \neq (0, 0)$ gives $\nu p - \mu q = [\nu\mu - \mu\nu, K] = 0$. $\square$

(In fact $[u, k] \mapsto u/4^k$ is an isomorphism $\Delta_A \cong \mathbb{Z}[1/4]$, with $\Delta_A^+ = \mathbb{Z}[1/4]_{\geq 0}$. We do not need this.)

Lemma 6 (Δ_B has rank at least two). *$x = [e_1, 0]$ and $y = [e_2, 0]$ are independent in Δ_B .*

Proof. $ax + by = [(a, b), 0]$. If this is $0 = [(0, 0), 0]$, then $(a, b)B^t = (0, 0)$ for some t . Now $vB = wB$ means $3v_1 + v_2 = 3w_1 + w_2$ and $v_1 + 3v_2 = w_1 + 3w_2$, hence $v = w$, since $\det B = 8 \neq 0$. By induction B^t is injective on $\mathbb{Z}^2$, so $(a, b) = (0, 0)$. $\square$

Theorem 7. *There is no injective additive map $\Delta_B \rightarrow \Delta_A$. In particular Δ_A and Δ_B are not isomorphic as groups. A fortiori $(\Delta_A, \Delta_A^+) \not\cong (\Delta_B, \Delta_B^+)$ as ordered groups, and the dimension triples are not isomorphic.*

Proof. Let f be additive and injective, and let x, y be as in Lemma 6. By Lemma 5 there are $(a, b) \neq (0, 0)$ with $af(x) + bf(y) = 0$, so $f(ax + by) = 0 = f(0)$, and hence $ax + by = 0$, contradicting Lemma 6. An isomorphism $\Delta_A \rightarrow \Delta_B$ would have an additive injective inverse. $\square$

Corollary 8. *Implication (1) is false for $A = [4]$ and $B = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$, under every reading of its hypothesis that holds when $\lambda_A = \lambda_B$, and for every notion of isomorphism (group, ordered group, dimension triple). Therefore Conjecture 00000009614 is false.*

5 Robustness

Eventual-range model. In the model of [1, §7.5] the argument is even more direct. $R_A = \mathbb{Q}$ and $\Delta_A = \{x \in \mathbb{Q} : 4^k x \in \mathbb{Z} \text{ for some } k\} = \mathbb{Z}[1/4]$. Any two elements $m/4^k$ and $m'/4^{k'}$ satisfy $m'4^k \cdot \frac{m}{4^k} - m4^{k'} \cdot \frac{m'}{4^{k'}} = 0$. Since $\det B = 8 \neq 0$, we have $R_B = \mathbb{Q}^2$ and $\mathbb{Z}^2 \subseteq \Delta_B \subseteq \mathbb{Q}^2$. So $\text{rank } \Delta_B = \dim_{\mathbb{Q}}(\Delta_B \otimes \mathbb{Q}) = 2 \neq 1 = \text{rank } \Delta_A$. Rank is an invariant of abstract abelian groups.

A second obstruction (dimension triples). By Krieger's theorem [1, §7.5], $(\Delta_A, \Delta_A^+, \delta_A) \cong (\Delta_B, \Delta_B^+, \delta_B)$ if and only if A and B are shift equivalent. Shift-equivalent matrices have the same nonzero spectrum [1, §7.4]. Here the nonzero spectra are $\{4\}$ and $\{4, 2\}$. Equivalently, $\text{tr } A^k = 4^k \neq 4^k + 2^k = \text{tr } B^k$, which counts the period- k points. This independent argument covers the triple reading. The rank argument covers all readings.

Mixing. Both matrices are positive, so the shifts are mixing under any presentation (edge shift; X_A is also the full 4-shift as a vertex shift).

Why the conjecture fails. The rank of Δ_M is the number of nonzero eigenvalues of M (with algebraic multiplicity). Any condition that depends only on λ_M ignores the non-Perron part of the spectrum. Our B has reducible characteristic polynomial. The conjecture places no restriction on the characteristic polynomial, and we do not claim anything about a variant restricted to irreducible characteristic polynomials.

6 Lean formalization

The directory `lean4/` is a self-contained Lean 4.19.0 project (core library only, no Mathlib). File `Main.lean`, namespace `DimGroup`:

- Vectors are `Fin n → Int` and matrices are `Fin n → Fin n → Int`. `rmul M v` is vM (row vector) and `pw M t v` is vM^t , for arbitrary n .
- `Rel M` is the relation of Definition 1. Its equivalence is proved, and `DG M` is the quotient type Δ_M . `Add`, `Neg` and `Zero` are defined on the quotient, and well-definedness is proved. `add_comm'`, `add_assoc'`, `zero_add'` and `neg_add'` prove the abelian group axioms. `Pos M` is the positive cone.
- `GroupIso` (additive bijection) and `OrderIso` (additionally $x \in \Delta^+ \iff f(x) \in \Delta^+$) are structures. `OrderIso.refl` shows these notions are not vacuous.
- `rank_one`: for every 1×1 matrix, any p, q satisfy $(a+1)p + (b+1)q = (c+1)p + (d+1)q$ for some $(a, b) \neq (c, d)$. This is the integer relation with its positive and negative parts separated, so that only positive multiples `nsml` are needed. `rank_two`: in Δ_B , such an identity for $[e_1, 0], [e_2, 0]$ forces $(a, b) = (c, d)$. `no_injective_hom`, `not_groupIso_AB` and `not_orderIso_AB` prove Theorem 7.
- `Primitive M` means $M \geq 0$ and some M^k , $k \geq 1$, is positive. `IsPerronEig M lam` means there is a positive v with $vM = \lambda v$. These are proved for A, B with $\lambda = 4$ (`primitive_A`, `perron_B`, ...). `second_eig_B` gives the eigenvalue 2, and `pathCount_A`, `pathCount_B` give the path counts 4^k and $2 \cdot 4^k$.

- **IfDirection Cond** is the statement “for all n, m and all $A \in \mathbb{Z}^{n \times n}$, $B \in \mathbb{Z}^{m \times m}$ primitive with integer Perron eigenvalues λ_A, λ_B satisfying $\text{Cond}(\lambda_A, \lambda_B)$, Δ_A and Δ_B are order-isomorphic”. This is a special case of the “if” direction. `conjecture_00000009614_false` proves $\neg \text{IfDirection Cond}$ for every reflexive **Cond**. `conjecture_00000009614_iff_false` refutes the biconditional, `conjecture_00000009614_group_false` refutes the version with mere group isomorphism, and `conjecture_00000009614_false.Q` instantiates the concrete reading $\lambda_A = \pm \lambda_B$ (field $\mathbb{Q}$, units $\{\pm 1\}$). **nonvacuous** records that all hypotheses are satisfiable.

Not formalized: the Perron–Frobenius theorem itself (that a positive eigenvector characterizes the spectral radius, used only to identify **IsPerronEig** with “Perron eigenvalue”; for B the full spectrum $\{4, 2\}$ is exhibited by eigenvectors in Lean), the identification of the direct limit with the eventual-range model, and Krieger’s theorem (used only in the second, redundant obstruction). **lake build** succeeds with no warnings. There is no **sorry**, no **native_decide**, and no added axiom. **#print axioms** shows only **propext** and **Quot.sound**.

7 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, core only
python3 verify.py          # Python 3 standard library
pdflatex report.tex
```

verify.py checks the same facts independently, with exact rational arithmetic. It computes the characteristic polynomials, the spectral radius (also by power iteration), and the path counts. It tests rank one and rank two in the eventual-range model, and runs a brute-force check of the identifications in the direct limit. It also verifies the trace and spectrum invariants.

References

- [1] D. Lind and B. Marcus, *An Introduction to Symbolic Dynamics and Coding*, Cambridge University Press, 1995 (§4.2–4.5 Perron–Frobenius theory and mixing; §7.4 shift equivalence; §7.5 dimension groups and Krieger’s theorem).
- [2] W. Krieger, On dimension functions and topological Markov chains, *Invent. Math.* 56 (1980), 239–250.